La Scuola d'Italia / Dr. Huson / IB Math 11
11 November 2026

Nickname:

3.1 Classwork: Sets and Venn Diagrams Review (no calculator)

$x \in A$	x is an element of A	$A \cap B$	intersection: in A *and* in B
$x \notin A$	x is not an element of A	$A \cup B$	union: in A *or* B (or both)
$n(A)$	the number of elements in A	A'	complement: not in A
$A \subseteq B$	A is a subset of B	U	the universal set
$\varnothing$	the empty set, $\{\ \}$	$\{x \mid \ldots\}$	the set of all x such that $\ldots$

1. A *prime number* is a positive integer with exactly two positive factors: 1 and itself. So 1 is *not* prime.

   Let $P = \{\text{prime numbers less than } 30\}$.

   (a) List the elements of P, and write down $n(P)$.

   (b) Write down the only even prime number. Explain why every other even number is not prime.

   (c) State whether each statement is true or false.

   (i) $9 \in P$ (ii) $2 \in P$ (iii) $1 \in P$ (iv) $\{3, 5\} \subseteq P$ (v) $27 \notin P$

   (d) Let $F = \{\text{positive factors of } 30\}$. List the elements of F, and find $F \cap P$. Describe $F \cap P$ in words.

   (e) Is the set of *all* prime numbers finite or infinite?

2. $\mathbb{N} = \{0, 1, 2, 3, \ldots\}$, $\mathbb{Z} = \{0, \pm 1, \pm 2, \ldots\}$, $\mathbb{Z}^+ = \{1, 2, 3, \ldots\}$, $\mathbb{Q}$ = rational numbers $\frac{p}{q}$ $(p, q \in \mathbb{Z},\ q \neq 0)$, $\mathbb{R}$ = real numbers.

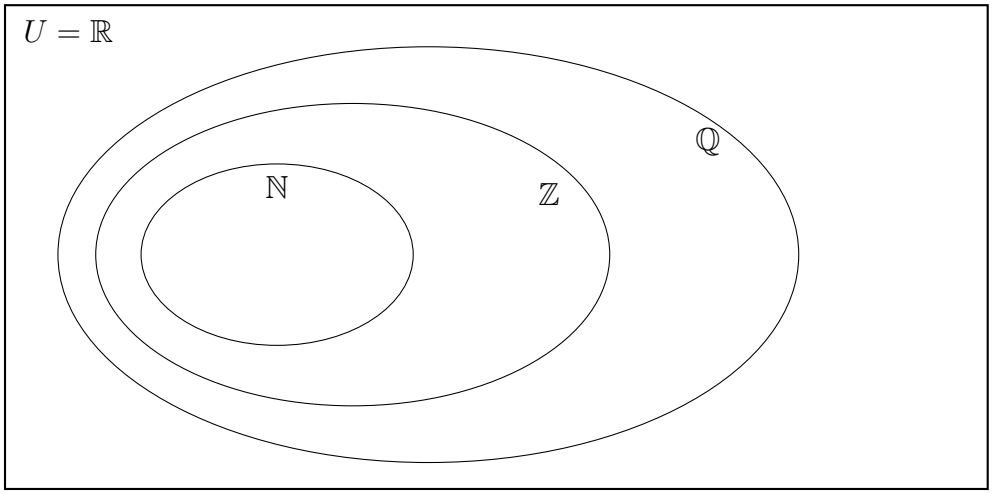
(a) State whether each statement is true or false.

(i) $-3 \in \mathbb{N}$ (ii) $0 \in \mathbb{N}$ (iii) $0 \in \mathbb{Z}^+$ (iv) $\sqrt{2} \in \mathbb{Q}$

(v) $0.25 \in \mathbb{Q}$ (vi) $0.\overline{3} \in \mathbb{Q}$ (vii) $\mathbb{Z} \subseteq \mathbb{Q}$ (viii) $\mathbb{Q} \subseteq \mathbb{Z}$

(b) Write each number in the correct region of the Venn diagram.

$$7, \quad -2, \quad 0, \quad \frac{1}{3}, \quad -0.6, \quad \sqrt{16}, \quad \sqrt{5}, \quad \pi$$



3. (a) List the elements of each set, and write down n of the set. If the elements cannot be listed, say why.

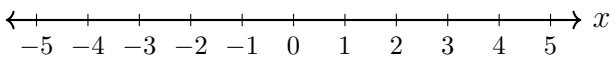
(i) $A = \{x \in \mathbb{Z} \mid -2 \leq x < 4\}$ (ii) $B = \{x \in \mathbb{N} \mid x < 5\}$

(iii) $C = \{x \in \mathbb{Z}^+ \mid x^2 < 20\}$ (iv) $D = \{x \in \mathbb{R} \mid 1 < x < 2\}$

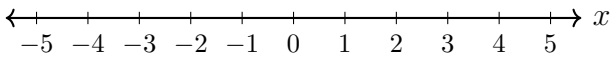
Nickname:

(b) Show each set on the number line.

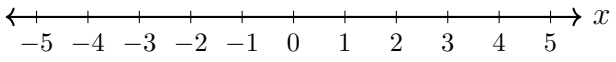
(i) $\{x \in \mathbb{R} \mid -1 < x \leq 3\}$



(ii) $\{x \in \mathbb{Z} \mid -3 < x \leq 1\}$

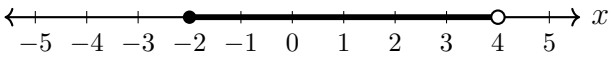


(iii) $\{x \mid x < 0\} \cup \{x \mid x \geq 2\}$

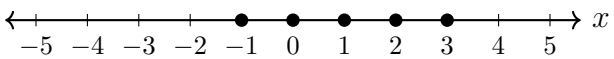


(c) Write each set in set-builder notation.

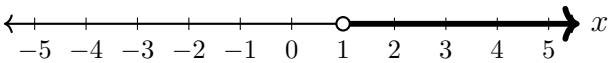
(i)



(ii)



(iii)

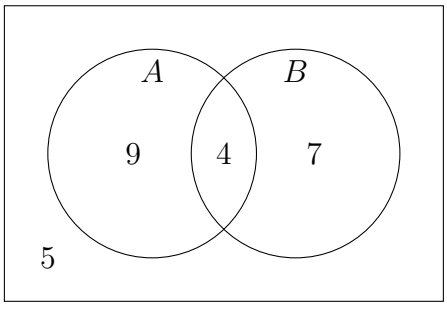


4. The numbers in the Venn diagram are the numbers of elements in each region. Find:

(a) $n(A)$ (b) $n(B')$

(c) $n(A \cup B)$ (d) $n(A \cap B')$

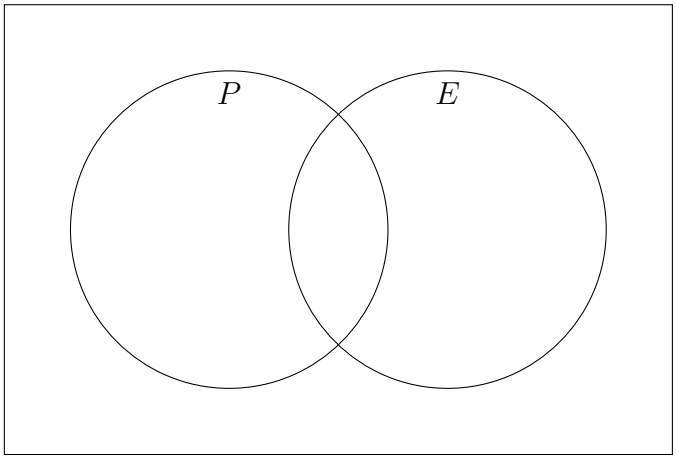
(e) $n\big((A \cap B)'\big)$ (f) $n(U)$



5. Let $U = \{x \in \mathbb{Z}^+ \mid x \leq 20\}$, $P = \{\text{prime numbers}\}$ and $E = \{\text{even numbers}\}$.

(a) List the elements of P and of E.

(b) Write all 20 elements of U in the correct regions of the Venn diagram.



(c) List the elements of

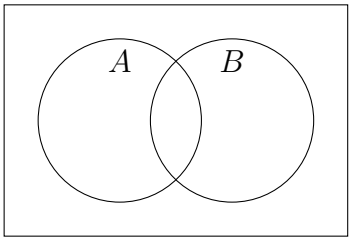
(i) $P \cap E$ (ii) $P \cap E'$ (iii) $(P \cup E)'$

(d) Find $n(P \cup E)$ by counting. Then show that $n(P \cup E) = n(P) + n(E) - n(P \cap E)$.

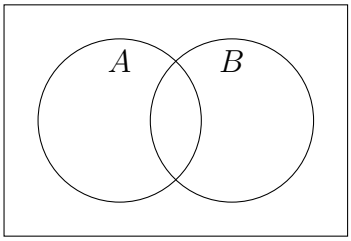
(e) A number is chosen at random from U. Find the probability that it is neither prime nor even.

Nickname:

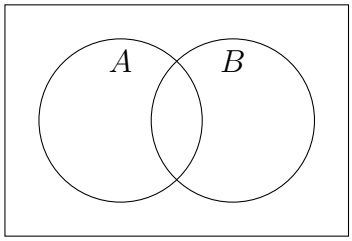
6. Shade the region described under each Venn diagram.



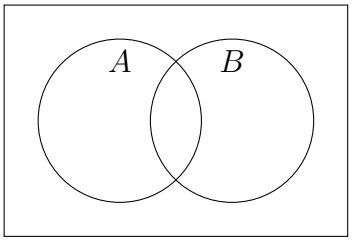
(a) $A \cap B'$



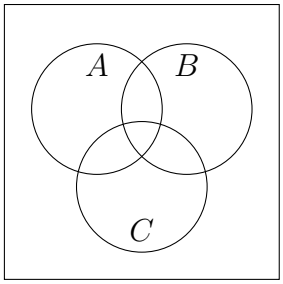
(b) $(A \cup B)'$



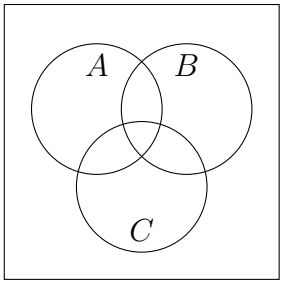
(c) $A' \cup B'$



(d) B'

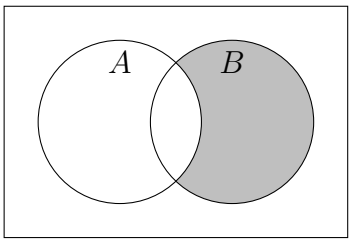


(e) $A \cap B \cap C$

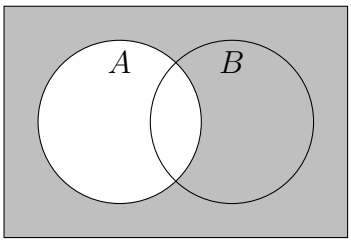


(f) $(A \cup B) \cap C'$

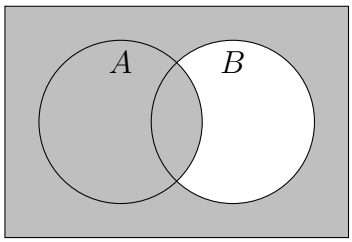
7. Write an expression, using $\cap$, $\cup$ and $'$, for each shaded region.



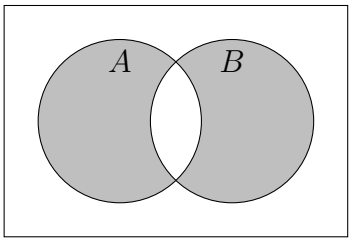
(a) _____



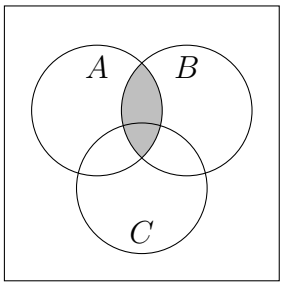
(b) _____



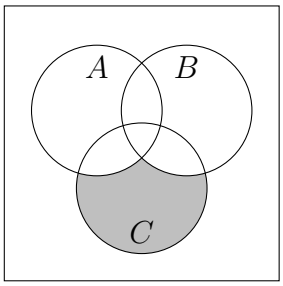
(c) _____



(d) _____



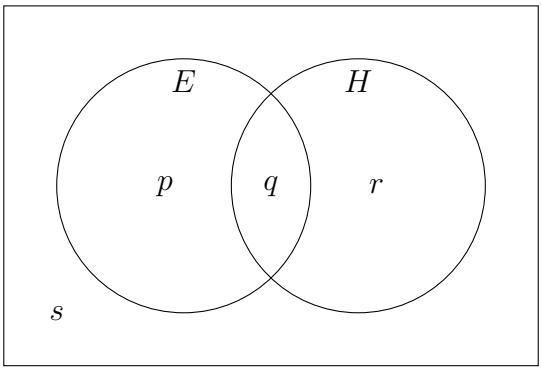
(e) _____



(f) _____

(g) Describe the region in diagram (d) in words.

8. In a class of 32 students, 19 study Economics (E) and 15 study History (H). Six students study neither subject. This information is shown in the Venn diagram, where p, q, r and s are numbers of students.



(a) Write down the value of s.

(b) Find the value of q.

(c) Find the value of p and the value of r.

(d) A student is chosen at random from the class. Find the probability that the student studies exactly one of the two subjects.

9. At a school, 65% of the students take music lessons (M) and 30% take dance classes (D). 15% of the students take neither.

(a) Draw a Venn diagram, and find the percentage of students who take both music and dance.

(b) A student is selected at random. Find the probability that the student

(i) takes dance but not music;

(ii) takes exactly one of music and dance.